

BACHELOR OF TECHNOLOGY (CBCS) (2021-COURSE)
B. Tech. Sem - V INFORMATION TECHNOLOGY : SUMMER : 2024
SUBJECT: COMPUTER ARCHITECTURE & ORGANIZATION

Day : Tuesday
Date : 14/05/2024

S-25757-2024

Time : 10:00 AM-01:00 PM
Max. Marks : 60

N.B.

- 1) All questions are **COMPULSORY**.
- 2) Figures to the **RIGHT** indicate **FULL** marks.
- 3) Assume suitable data **WHEREVER** necessary.
- 4) Draw neat labeled diagrams **WHEREVER** necessary.

Q.1 What are the latest trends in the field of computer architecture? Give pointwise description. (10)

OR

Q.1 Write note on the following: (10)
a) Cloud Computing b) Quantum Computers

Q.2 Describe the different features of VHDL. (10)

OR

Q.2 Write a VHDL code for half-adder using dataflow and behavioral methods. Mention important comments. (10)

Q.3 Explain the hardwired control method design using state-table method. (10)

OR

Q.3 List and explain the micro-operations for the execution of following instruction: (10)
ADD AL, [DI]

Q.4 Explain the working of 3D NAND memory with suitable diagram. Also give its applications. (10)

OR

Q.4 Explain the cache read and write operations with suitable diagrams. (10)

Q.5 Explain the supercomputer architecture with neat diagram. (10)

OR

Q.5 Describe the shared memory architecture with suitable diagram. (10)

Q.6 What are the different types of hazards? Explain with suitable examples. (10)

OR

Q.6 What is out-of-order execution? Explain with suitable example. (10)
